

1 Topological Recurrence in EEG Dynamics:
2 A Complementary, Wake-to-REM-Graded Descriptor
3 of Sleep-Stage Organisation

4 **Short title:** Topological recurrence across sleep stages

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Abstract

Background. Sleep-stage EEG analysis conventionally relies on spectral and complexity measures. Whether the geometric organisation of neural dynamics provides complementary information remains untested. Here I test whether a topological recurrence descriptor derived from persistent homology indexes a dimension of sleep-stage EEG organisation that is statistically distinct from spectral and complexity measures.

New method. K_0 , a within-subject z-scored summary of persistent homology of time-delay embeddings ($m=10$, $\tau=2$) of single-channel frontal-midline (Fpz–Cz) EEG, was computed on 197 nights from 100 subjects (Sleep-EDF Expanded, a public polysomnography corpus). Mixed-effects models, binomial generalised linear models (GLMs), and leave-one-subject-out classification benchmarked K_0 against band power, spectral entropy, permutation entropy, and Lempel–Ziv complexity.

Results. K_0 , varying systematically across sleep stages, is statistically distinct from spectral content (Pearson r -0.49 to -0.25 with log band powers; opposite rapid eye movement (REM)–Wake direction), with construct validity supported by independent-cohort replication, twofold lifespan attenuation, and within-subject benzodiazepine invariance. The REM–Wake separation is large ($+1.01$ standard deviation (SD); $p = 2.62 \times 10^{-182}$) and preserved across embeddings.

Comparison with existing methods. A binomial GLM confirms K_0 carries information statistically independent of band power, especially for REM vs active-ocular wake (Δ Akaike Information Criterion (AIC) $= -744.6$, odds ratio (OR) $= 5.04$). Leave-one-subject-out area under the receiver operating characteristic curve (AUC) values are 0.79 (K_0) and 0.91 (band power); combining adds a small gain (Δ AUC $\approx +0.004$, paired-bootstrap $p < 10^{-3}$).

Conclusions. K_0 is a topologically distinct, spectrally non-redundant descriptor of sleep-stage EEG organisation.

Keywords: persistent homology; topological data analysis; sleep EEG; REM sleep; state-space reconstruction; Sleep-EDF.

1 Introduction

Electroencephalography (EEG) has long provided the foundation for classifying sleep stages, traditionally distinguishing wakefulness, non-rapid eye movement (NREM) sleep, and rapid eye movement (REM) sleep through spectral power, oscillatory features, and statistical descriptors [1, 2]. Deep NREM sleep is characterised by high-amplitude slow oscillations, whereas REM sleep exhibits low-amplitude, broadband activity that closely resembles quiet wakefulness. Despite this apparent similarity, REM sleep differs profoundly from wakefulness in phenomenology, supporting vivid, internally generated experiences that are largely decoupled from external sensory input [3].

This dissociation presents a challenge for conventional EEG analysis. Standard measures reliably separate deep sleep from lighter stages via delta power, but often struggle to distinguish REM sleep from wakefulness due to their shared desynchronised characteristics, particularly when limited to single-channel data [2, 4]. This limitation suggests that commonly used EEG metrics may primarily reflect arousal, synchronisation, or spectral composition, rather than deeper organisational features of neural dynamics relevant to internally generated activity.

Converging evidence from neurophysiology indicates that REM sleep is associated with distinctive thalamocortical and neuromodulatory configurations. During REM sleep, thalamic relay nuclei reduce the transmission of external sensory input while cortico-cortical and intrathalamic activity remain active, effectively gating the cortex from the external environment [2, 5]. At the same time, REM sleep is characterised by elevated cholinergic tone and reduced monoaminergic drive, a neuromodulatory profile thought to promote sustained cortical activation and recurrent intracortical interactions [6, 7]. These mechanistic considerations form the speculative interpretive context in which I situate the present empirical findings; I do not test them directly here.

From a computational perspective, such brain states may differ not primarily in spectral content but in the organisation of their underlying dynamics. Dynamical systems approaches provide tools for examining this organisation directly. Time-delay embedding offers a principled method for reconstructing the state space of an underlying dynamical system from scalar time series data, revealing geometric structure that is not apparent in the time or frequency domains [8]. Persistent homology has been applied to neuroimaging time series in functional MRI, MEG, and EEG to characterise topological structure in brain dynamics [9–13], but applications to single-channel polysomnography EEG and to the discrimination of REM sleep specifically are limited. Within this framework, recurrence reflects the tendency of a system’s trajectory to revisit similar regions of state space over time. Importantly, recurrence need not correspond to simple periodicity but can arise from distributed feedback loops operating across multiple timescales.

Topological data analysis (TDA) provides a complementary framework for quantifying such recurrence in a scale-invariant manner. Persistent homology characterises the appearance and persistence of topological features—such as loops—across a range of spatial scales [14, 15]. Applied to delay-embedded neural data, first-homology features capture loop-like structures corresponding to recurrent trajectories in reconstructed state space. These features provide a natural

summary of distributed recurrence without reliance on predefined frequency bands or oscillatory templates.

Unlike prior applications of persistent homology to neuroimaging — which have operated on functional-connectivity networks in fMRI [9, 12], in MEG [10], or in coupled time-series networks [11], and on Mapper-based spatiotemporal neuroimaging representations [13] — the present study applies persistent homology directly to a delay-embedded state-space reconstruction from a single EEG channel, on a polysomnographic-scale sleep corpus, in a within-subject normalised framework, and benchmarks the resulting recurrence metric head-to-head against the spectral, entropy, and algorithmic-complexity measures already used in clinical sleep staging. The framing of the present work is one of construct validity rather than of discrimination: I test whether a recurrence-based descriptor (i) varies systematically across sleep stages and replicates in an independent cohort, (ii) is statistically independent of conventional spectral and complexity measures, (iii) is robust to embedding-parameter choice, (iv) attenuates with age in a manner consistent with documented age-related changes in REM organisation, and (v) is invariant under a within-subject pharmacological challenge. The size of the REM-versus-Wake contrast on the metric is reported as one piece of evidence in this construct-validity argument rather than as a standalone classification claim.

This paper is intentionally empirical, descriptive and computational in scope. The Sleep-EDF Expanded corpus is observational; the present analyses characterise statistical associations between EEG dynamics and expert-scored sleep stages, and I make no causal claims about the neural mechanisms generating those associations. While the observed patterns are compatible with broader theoretical frameworks that emphasise internally closed organisational structure in conscious states, I restrict my claims here to reproducible distinctions in neural dynamics observable in human EEG.

2 Materials and Methods

2.1 Dataset

I analysed the Sleep-EDF Expanded dataset, a publicly available collection of overnight polysomnographic recordings from healthy adult participants [16]. A total of 197 nights from 100 unique subjects were included in the analysis, distributed across two recording sub-cohorts. The Sleep-Cassette sub-cohort contributed 153 nights from 78 subjects (age 25–101 years, mean 58.8 ± 22.1 SD, median 57; 41 female / 37 male; recordings approximately 24 h continuous, with no pharmacological manipulation). The Sleep-Telemetry sub-cohort contributed 44 nights from 22 subjects (age 18–79 years, mean 40.2 ± 18.1 SD, median 35; 15 female / 7 male) and additionally implemented a within-subject randomised Temazepam-versus-placebo crossover, with each participant contributing exactly one placebo night and one Temazepam night, allowing the headline contrast to be evaluated as a pharmacological-control sensitivity (see Demographic stratification below). Across both sub-cohorts the analysis cohort spanned 18–101 years of age (median 54, mean 54.7 ± 22.6 SD) and was 56% female (56 female / 44 male), broadly representative of the adult lifespan. Sex is reported as recorded in the original Sleep-EDF Expanded subject documentation and refers to sex assigned at birth; the dataset does not contain gender-identity

information. Each recording contains continuous EEG data together with expert-annotated hypnograms scored at 30-second resolution according to standard American Academy of Sleep Medicine (AASM) criteria [17].

Only standard sleep stages were included: wake (W), stage N1, stage N2, stage N3 (merging stages 3 and 4 in accordance with modern AASM standards [17]), and REM sleep. Epochs labelled as movement time or unscored were excluded. All analyses were conducted in accordance with the dataset’s original ethical approvals and usage guidelines.

2.2 EEG preprocessing

For each night, a single EEG channel was used to ensure consistency across recordings and to assess whether organisational structure could be detected under minimal spatial information. The Fpz–Cz derivation was available and used in every recording in the 197-night analysis cohort; no recordings required a fallback channel.

EEG signals were band-pass filtered between 0.5 and 40 Hz using a fourth-order Butterworth filter applied in a zero-phase manner (forward–backward filtering) to remove slow drift and high-frequency noise while avoiding phase distortion. Signals were subsequently downsampled to 50 Hz to reduce computational load while preserving temporal structure relevant to state-space reconstruction. Data were segmented into non-overlapping 30-second epochs, each aligned with the corresponding hypnogram annotation by sampling the sleep stage at the midpoint of the epoch.

2.3 State-space reconstruction

Each EEG epoch was transformed into a trajectory in reconstructed state space using time-delay embedding [8]. Given a univariate EEG signal $x(t)$, vectors were constructed as

$$X_t = [x(t), x(t + \tau), x(t + 2\tau), \dots, x(t + (m - 1)\tau)], \quad (1)$$

where m is the embedding dimension and τ is the delay in samples. Unless otherwise stated, analyses were conducted using embedding dimension $m = 10$ and delay $\tau = 2$ samples (corresponding to 40 ms at the resampled rate). This choice balances temporal resolution with sufficient dimensionality to capture recurrent structure and is consistent with prior work on EEG state-space reconstruction.

To reduce computational burden during topological analysis, embedded trajectories were subsampled by a factor of two within each epoch prior to persistent homology computation, yielding an effective sampling rate of 25 Hz for the reconstructed state-space point clouds. This step preserves large-scale geometric structure while limiting point-cloud size.

Each embedded point cloud was standardised by removing the mean and scaling to unit variance along each dimension. A small constant (10^{-8}) was added to the standard deviation solely to ensure numerical stability in cases of near-zero variance and does not materially affect the resulting geometry. This standardisation ensures that subsequent geometric analysis reflects structure rather than amplitude.

Epochs yielding fewer than 20 embedded points prior to within-epoch subsampling were excluded to ensure reliable estimation of topological features.

Time-delay embedding was performed using only time indices for which all delayed samples were defined; points requiring samples beyond the epoch boundary were excluded, avoiding the need for padding or extrapolation and preventing edge-related artefacts.

2.4 Persistent homology

To characterise the geometric and topological organisation of reconstructed EEG state spaces, I applied persistent homology to the delay-embedded point clouds derived from each 30-second epoch. Persistent homology provides a multiscale description of topological structure by quantifying features that remain stable across a range of spatial resolutions, thereby separating meaningful geometric organisation from sampling noise [14, 15].

For each epoch, the embedded state vectors were treated as a finite metric space and analysed using a Vietoris–Rips filtration. This procedure yields persistence diagrams for low-dimensional homology, summarising how topological features emerge and disappear as the spatial scale parameter is varied.

2.4.1 Vietoris–Rips filtration construction

Let $X = \{x_1, \dots, x_n\} \subset \mathbb{R}^m$ denote the set of delay-embedded vectors for a single epoch after standardisation and within-epoch subsampling as described in State-space reconstruction. The point cloud X was treated as a finite metric space under the Euclidean distance

$$d(x_i, x_j) = \|x_i - x_j\|_2. \quad (2)$$

Persistent homology was computed using a Vietoris–Rips filtration, which constructs a nested family of simplicial complexes parameterised by a scale parameter $\varepsilon \geq 0$. For a given scale ε , the Vietoris–Rips complex $\text{VR}_\varepsilon(X)$ contains a vertex for each point in X , an edge between vertices x_i and x_j whenever $d(x_i, x_j) \leq \varepsilon$, and a k -simplex whenever all pairwise distances between its $k + 1$ vertices are less than or equal to ε . As ε increases, simplices are added monotonically, yielding a filtration

$$\text{VR}_{\varepsilon_1}(X) \subseteq \text{VR}_{\varepsilon_2}(X) \subseteq \dots \quad \text{for } \varepsilon_1 \leq \varepsilon_2. \quad (3)$$

Following construction of the filtration, persistent homology was computed by tracking the birth and death of topological features as a function of the scale parameter ε . For each homology dimension k , this procedure yields a persistence diagram

$$\mathcal{D}_k = \{(b_i, d_i)\}, \quad (4)$$

where b_i and d_i denote the birth and death scales of the i -th topological feature, respectively. Features with infinite death times ($d_i = \infty$) correspond to structures persisting across all scales and were excluded from subsequent summaries, focusing analysis on transient recurrent features.

Consistent with prior work on recurrence structure in delay-embedded neural signals, analyses focused on first-order homology ($k = 1$), which captures loop-like features corresponding to

recurrent trajectories in reconstructed state space. Rather than analysing persistence diagrams directly, their global structure was summarised using total persistence, defined as

$$\text{TP}_k = \sum_{i:d_i < \infty} (d_i - b_i). \quad (5)$$

2.4.2 Definition of the topological recurrence metric K_0

Total persistence of H_1 features (TP_1) formed the basis of the primary topological recurrence measure. For notational convenience, I define

$$L_1 \equiv \text{TP}_1. \quad (6)$$

To isolate recurrence-based organisation without introducing arbitrary parameters, a primary topological recurrence metric K_0 was defined as

$$K_0 = \text{Norm}(L_1), \quad (7)$$

where $\text{Norm}(\cdot)$ denotes within-subject normalisation, implemented as z-scoring across all epochs for a given subject. This procedure removes individual baseline differences in signal scale and noise characteristics, placing all subjects on a comparable dimensionless scale while preserving within-subject ordering across sleep stages.

Crucially, K_0 contains no learned parameters; the embedding (m, τ) , the preprocessing settings (band-pass cutoffs, resample rate, epoch length), the within-epoch subsampling, the per-stage epoch sampling cap, and the random seed are all fixed by convention rather than fitted to data, and are shown to be robust across the tested embedding and preprocessing grids (Fig. 1; Supplementary S2, S3). It represents a direct operationalisation of scale-integrated recurrence in reconstructed neural state space and serves as the primary dependent variable in inferential analyses. Raw H_1 total persistence values are reported descriptively, while statistical testing is conducted exclusively on the normalised metric K_0 .

2.5 Statistical analysis

For each subject, K_0 values were averaged within each sleep stage to avoid pseudoreplication. All inferential analyses were conducted using within-subject contrasts, focusing on state-dependent changes rather than inter-individual variability.

Paired within-subject effect sizes were quantified using Cohen's d . In addition, linear mixed-effects models were fitted with sleep stage as a fixed effect and subject as a random intercept to assess overall stage effects and planned contrasts. Linear mixed-effects models were specified with sleep stage entered as a categorical fixed effect using treatment (dummy) coding, with wakefulness as the reference level, and subject included as a random intercept. Concretely, for subject s , epoch i , and sleep stage $c \in \{\text{W}, \text{N1}, \text{N2}, \text{N3}, \text{REM}\}$,

$$K_{0,i,s,c} = \beta_0 + \sum_{c \neq \text{W}} \beta_c \mathbb{I}[\text{stage}_{i,s} = c] + u_s + \varepsilon_{i,s,c}, \quad u_s \sim \mathcal{N}(0, \sigma_u^2), \quad \varepsilon_{i,s,c} \sim \mathcal{N}(0, \sigma_\varepsilon^2), \quad (8)$$

with β_0 the wake (reference) intercept, β_c the treatment-coded stage contrasts (REM, N1, N2, N3 versus W), u_s a subject random intercept, and $\varepsilon_{i,s,c}$ residual error. Planned contrasts between sleep stages were evaluated using linear hypothesis tests on the fitted β_c parameters. No random slopes were included, reflecting the limited number of observations per condition and the focus on population-level stage effects. Family-wise error was controlled using Holm correction where applicable. Sensitivity power analysis was performed to formally evaluate the adequacy of the analysis cohort against the headline contrast: the minimum detectable effect (MDE) at 80% power and $\alpha = 0.05$ (two-sided) was computed for a paired-samples t -test on subject-level REM-versus-Wake K_0 means using both the closed-form normal approximation

$$\text{MDE}_{d_z} = \frac{z_{1-\alpha/2} + z_{1-\beta}}{\sqrt{n}}, \quad (9)$$

and the exact non-central t distribution (`statsmodels.stats.power.TTestPower`). The same script computes the observed paired- t d_z on the headline contrast directly from the per-epoch K_0 data, for six cohort specifications. The minimum detectable paired effect-size was $d_z = 0.20$ for the per-night specification ($n = 197$ recordings); $d_z = 0.28$ for the full unique-subject cohort ($n = 100$, one night per person, Telemetry contributing its placebo night); $d_z = 0.32$ for Sleep-Cassette unique subjects ($n = 78$); and $d_z = 0.63$ for Sleep-Telemetry unique subjects ($n = 22$). The corresponding observed paired- t effect sizes were $d_z = 1.93$, $d_z = 1.92$, $d_z = 2.17$, and $d_z = 1.47$ respectively, between **2.4** $\times$ (telemetry-subjects) and **9.6** $\times$ (per-night) the MDE across all six cohort specifications. The cohort is therefore not underpowered for the headline effect: the analysis would have detected effects an order of magnitude smaller than the observed one. This calculation, together with the bootstrap confidence intervals and the independent-cohort replication reported below, addresses the sample-size concern by three independent lines of evidence.

2.6 Computational implementation and reproducibility

All analyses were performed in Python (version ≥ 3.10). EEG preprocessing and epoching were implemented using NumPy and SciPy [18, 19]. Time-delay embedding was implemented using direct array construction without dimensionality reduction.

Persistent homology was computed using the Ripser library [20], which implements an efficient cohomology-based algorithm with clearing optimisation for Vietoris–Rips filtrations. The maximum homology dimension was set to 1 (i.e., H_0 and H_1 features); all other Ripser parameters were left at the library defaults.

To control computational load while preserving representative sampling, up to 30 epochs per sleep stage were randomly selected per night for topological analysis. To ensure reproducibility, random epoch sampling was performed using a fixed random seed (seed = 0 in NumPy’s default random number generator). Summary statistics including feature count, total persistence, and maximum persistence were computed for each homology dimension; only total persistence was used for inferential analyses.

Linear mixed-effects models were fitted using the `statsmodels` package [21]. All analysis scripts, parameter settings, and instructions required to reproduce the results are publicly avail-

able; see the Code Availability section at the end of the paper for repository and DOI details.

For comparator EEG metrics (spectral entropy, permutation entropy, Lempel–Ziv complexity, log band power), the following implementation choices were used to ensure a fair comparison with K_0 . Power spectral density was estimated by Welch’s method with segment length $n_{\text{fft}} = \min(2048, n_{\text{epoch}})$ over the frequency range 0.5–40 Hz on the same 30-second, 50 Hz, 0.5–40 Hz band-pass filtered epochs used for K_0 . Spectral entropy was computed as the normalised Shannon entropy of the resulting PSD,

$$H_{\text{spec}} = -\frac{1}{\log K} \sum_{k=1}^K p_k \log p_k, \quad p_k = \frac{P(f_k)}{\sum_{k'=1}^K P(f_{k'})}, \quad (10)$$

where $P(f_k)$ denotes the Welch PSD estimate at frequency bin f_k and K is the number of bins inside the 0.5–40 Hz band. Log band powers were obtained by trapezoidal integration of the PSD over the canonical bands (δ 0.5–4, θ 4–8, α 8–12, σ 12–15, β 15–30 Hz). Permutation entropy at embedding order m and delay $\tau_{\text{PE}} = 1$ sample was computed from the relative frequencies $p(\pi)$ of the $m!$ ordinal patterns and normalised to the unit interval,

$$H_{\text{PE}}(m) = -\frac{1}{\log(m!)} \sum_{\pi \in S_m} p(\pi) \log p(\pi), \quad (11)$$

where S_m denotes the set of $m!$ ordinal patterns at order m (used here to label patterns, not to invoke the symmetric group structure), with $m = 5$ in the primary analysis (sensitivity over $m \in \{3, 4, 5, 6\}$ reported in Supplementary S5). Lempel–Ziv complexity was computed on the median-thresholded binary string $b_{1:n}$ of the epoch using the LZ76 production count $c(n)$ and the standard normalisation

$$C_{\text{LZ}} = c(n) \frac{\log_2(n)}{n}, \quad (12)$$

yielding a roughly $[0, 1]$ -bounded complexity score. All comparator metrics were therefore computed on identical epoch material and within-subject z-scored prior to inferential analysis, matching the procedure applied to K_0 . All methodological choices described above correspond directly to the released analysis code, ensuring full alignment between implementation and reported methods.

2.7 Wake-quality control and wake-subclass labelling

To address concerns about possible eye-movement contamination of frontal EEG during wake, each wake epoch was characterised by a panel of quality-control features computed on the co-recorded electrooculogram (EOG horizontal) and EMG (submental): low-frequency EOG RMS (0.3–5 Hz), the number of EOG peaks exceeding 2.5σ within the epoch, the absolute EEG–EOG correlation, the alpha ratio (8–12 Hz over 1–20 Hz), the variance of low-pass filtered Fpz–Cz, peak-to-peak amplitudes, and EMG RMS. Per-recording quartile thresholds and a 6-criterion “active” score were used to assign each wake epoch to one of three subclasses— W_{quiet} , $W_{\text{active ocular}}$, or W_{bad} —following the procedure of recording-percentile thresholding (Track 2). A complementary per-subject MAD-based quality-control table (Track 3) supported the artefact-control analyses described below.

2.8 Artefact control: EOG-corrected EEG and EOG-channel negative control

To verify that the topological recurrence effect was carried by the EEG signal rather than by ocular bleed-through, the following control analyses were added. For each recording, an EOG-corrected EEG channel was constructed by full-recording linear regression of the EOG horizontal channel onto Fpz–Cz: the ordinary-least-squares slope

$$\hat{\beta} = \frac{\text{cov}(\text{EEG}, \text{EOG})}{\text{var}(\text{EOG})} \quad (13)$$

was estimated once per recording on the full continuous trace and subtracted to yield the corrected channel

$$\text{EEG}_{\text{corrected}}(t) = \text{EEG}(t) - \hat{\beta} \text{EOG}(t). \quad (14)$$

H_1 total persistence and K_0 were then recomputed on (i) the original raw EEG, (ii) the EOG-corrected EEG, and (iii) the EOG horizontal channel itself, restricted to REM and the wake subclasses. The EOG-channel analysis serves as a negative control: if the K_0 effect were primarily driven by ocular activity, it should be at least as large on the EOG channel as on the EEG channels.

2.9 Multi-channel control

The headline analysis was repeated on the EEG Pz–Oz derivation (where present) using identical preprocessing and embedding parameters. This provides a posterior-electrode control to test whether the recurrence effect is anatomically localised to frontal midline electrodes or generalises across cortex.

2.10 Independent-cohort replication

Sleep-EDF Expanded contains two independent recording cohorts: Sleep-Cassette ($n = 153$ paired nights, healthy adults) and Sleep-Telemetry ($n = 44$ paired nights, an independent sample including subjects with mild sleep complaints). Recordings were assigned to cohort by filename prefix (SC* versus ST*), and the headline planned contrasts on K_0 and on each baseline EEG metric were refit separately within each cohort. This provides an independent-cohort replication test for the central claims.

2.11 All-pairwise stage contrasts and monotonicity

To explicitly characterise the gradient of K_0 across the five sleep stages, all ten pairwise contrasts (W, N1, N2, N3, REM) were estimated from the same mixed-effects model used for the headline planned contrasts, with Holm correction across the family. Monotonicity was assessed as a two-level test, computed per recording (one ρ per night, $n = 197$ nights from 100 subjects). For each night r , the Spearman rank correlation between the recording-level mean K_0 in each stage and the ordinal rank $\text{rk}(c)$ assigned to stage c along the sequence $W \rightarrow N1 \rightarrow N2 \rightarrow N3 \rightarrow \text{REM}$

321 was computed,

$$\rho_r = \text{Spearman}(\bar{K}_{0,r,c}, \text{rk}(c)), \quad \text{rk}(W) = 1, \text{rk}(N1) = 2, \dots, \text{rk}(\text{REM}) = 5. \quad (15)$$

322 The set $\{\rho_r\}_{r=1}^n$ was then tested against zero using a one-sample t -test,

$$t = \frac{\bar{\rho}}{s_\rho/\sqrt{n}}, \quad (16)$$

323 with $\bar{\rho}$ and s_ρ the sample mean and standard deviation of the per-recording Spearman correla-
324 tions.

325 2.12 Robustness and sensitivity analyses

326 To verify the robustness of the headline contrast, six sensitivity analyses were performed; full
327 methods and results for each are reported in the Supplementary Materials. (i) A subject-level
328 non-parametric bootstrap (1,000 resamples) gave 95% percentile CIs on the headline planned
329 contrasts (Supplementary S6). (ii) A 4×3 preprocessing-sensitivity grid (band-pass cutoffs
330 $\times$ resample rate) was evaluated on a 30-night random subsample (Supplementary S2). (iii)
331 Embedding-parameter diagnostics (AMI(τ) and FNN(m); Kennel et al. 1992) were computed
332 on a 30-recording subset, and the headline contrast was additionally re-fitted at three diagnostic-
333 favoured embedding cells outside the original robustness grid ($m \in \{6, 10, 12\}$, $\tau = 11$; Sup-
334plementary S3). (iv) Statistical assumption diagnostics on the headline mixed-effects model
335 (Shapiro–Wilk on residuals, Levene’s homoscedasticity, residual skewness, excess kurtosis; Sup-
336plementary S1). (v) Permutation entropy was recomputed at orders 3–6 (delay = 1) on every
337 Fpz–Cz epoch and the comparator REM-versus-Wake contrast was refitted at each order (Sup-
338plementary S5). (vi) An ICA-based ocular-artefact sensitivity analysis was performed on a
339 20-subject subset, fitting FastICA on the two EEG channels with EOG horizontal as a reference
340 signal for `find_bads_eog` (Supplementary S4).

341 2.13 Incremental binomial GLM: does K_0 add information beyond band 342 power?

343 To directly test whether K_0 contributes information beyond conventional spectral features, I
344 fitted four binomial generalised linear models with subject fixed effects on per-epoch data, sep-
345 arately for the contrasts REM versus W_{quiet} and REM versus $W_{\text{active ocular}}$. Let $y_{i,s} \in \{0, 1\}$
346 indicate REM ($y = 1$) versus the relevant wake subclass ($y = 0$) for epoch i in subject s . Models
347 took the form

$$\text{logit Pr}(y_{i,s} = 1) = \alpha_s + \sum_{b \in \{\delta, \theta, \alpha, \beta\}} \gamma_b \log P_{b,i,s} + \kappa K_{0,i,s}, \quad (17)$$

348 where α_s is a subject fixed effect, $\log P_{b,i,s}$ is the log power in band b (with b indexing the
349 canonical bands $\delta, \theta, \alpha, \beta$), and κ is the K_0 coefficient. Model A omitted the K_0 term ($\kappa \equiv 0$);
350 Model B included it; Model C omitted the band-power terms ($\gamma_b \equiv 0$); Model D was identical

351 to Model B. I report the likelihood-ratio test of A versus B,

$$\Lambda = -2(\ell_A - \ell_B) \sim \chi_1^2 \quad \text{under } H_0 : \kappa = 0, \quad (18)$$

352 where ℓ_A and ℓ_B are the maximised log-likelihoods of Models A and B respectively; the change
353 in Akaike Information Criterion

$$\Delta\text{AIC} = \text{AIC}_B - \text{AIC}_A = 2 - \Lambda, \quad (19)$$

354 and the odds ratio $\text{OR} = \exp(\hat{\kappa})$ for K_0 in Model B. I note that these binomial GLMs treat
355 30-second epochs as conditionally independent given the subject fixed effects: the reported
356 standard errors and likelihood-ratio statistics do not account for within-subject autocorrelation
357 in successive epochs, and the magnitude of the AIC improvement (and hence the size of ΔAIC)
358 scales with the sample size used to fit the models, so absolute ΔAIC values should be interpreted
359 in light of the per-contrast epoch counts rather than as cohort-invariant quantities.

360 2.14 Leave-one-subject-out classification

361 To translate the inferential findings into a predictive performance estimate, I performed leave-
362 one-subject-out classification using `scikit-learn`. For each contrast (REM versus W; REM
363 versus NREM; REM versus all other stages), three feature sets (i) K_0 alone, (ii) four log band
364 powers (delta, theta, alpha, beta), (iii) the combined set, were trained as logistic regression
365 and random-forest classifiers on all but one subject, evaluated on the held-out subject, and
366 the area under the receiver operating characteristic curve (AUC), balanced accuracy, sensitivity
367 and specificity were averaged across folds. The combined-versus-bandpower AUC difference
368 was tested by a paired non-parametric bootstrap over held-out subjects. For each of $B = 1,000$
369 bootstrap iterations, the subject pool was resampled with replacement and the pooled-prediction
370 AUCs for the combined and band-power feature sets were recomputed on the resampled scores,
371 yielding paired statistics

$$\Delta\text{AUC}^{(b)} = \text{AUC}_{\text{combined}}^{(b)} - \text{AUC}_{\text{bp}}^{(b)}, \quad b = 1, \dots, B. \quad (20)$$

372 The reported point estimate is the mean of $\{\Delta\text{AUC}^{(b)}\}_{b=1}^B$, the 95% interval is the [2.5, 97.5]
373 percentile, and the one-sided p -value is the fraction of bootstrap replicates with $\Delta\text{AUC}^{(b)} \leq 0$.

374 2.15 Regime-specific classification

375 To resolve whether the pooled REM-versus-Wake AUC reflects a uniform discrimination or
376 hides regime-specific structure, leave-one-subject-out logistic regression was repeated separately
377 for two narrower contrasts: REM versus quiet wake (W_{quiet}) and REM versus active-ocular wake
378 ($W_{\text{active ocular}}$), using the wake-subclass labels described in the Wake-quality control subsection.
379 Per-test-epoch held-out scores were stored and used to compute pooled AUCs together with
380 subject-level non-parametric bootstrap 95% confidence intervals (1,000 iterations resampling
381 held-out subjects with replacement) and a paired-bootstrap test for combined-versus-bandpower
382 AUC differences in each regime. As a second test of the cortical specificity of band-power's

discrimination, the band-power LOSO was also re-run on EOG-regressed Fpz-Cz (using the per-recording linear regression coefficients computed by the main pipeline).

2.16 Demographic stratification: lifespan, sex, and pharmacological control

Demographic information was extracted from the Sleep-EDF Expanded subject spreadsheets (SC-subjects.xls and ST-subjects.xls). The 100 unique subjects were classified by age group (18–29, 30–44, 45–59, 60–74, 75+) and sex (F/M); the REM-versus-Wake mixed-LM planned contrast on K_0 was refitted within each (age-group, sex) cell on a one-night-per-subject sample (Telemetry contributed its placebo night to avoid drug-protocol confounds). For the Telemetry sub-cohort separately, the within-subject Temazepam-versus-placebo crossover was used as a pharmacological control: each subject contributed exactly one placebo and one Temazepam night, allowing the REM-versus-Wake contrast under each drug condition to be estimated and compared. The stage $\times$ drug model retained a subject random intercept and treated stage c and the binary drug indicator (Temazepam vs the placebo reference) as crossed categorical fixed effects with their interaction. Writing $I_c = \mathbb{1}[\text{stage} = c]$ for the stage indicators and $I_T = \mathbb{1}[\text{drug} = \text{Temazepam}]$ for the drug indicator,

$$K_{0,i,s,c} = \beta_0 + \sum_{c \neq W} \beta_c I_c + \beta_T I_T + \sum_{c \neq W} \beta_{cT} I_c I_T + u_s + \varepsilon_{i,s,c}, \quad (21)$$

with $u_s \sim \mathcal{N}(0, \sigma_u^2)$ and $\varepsilon_{i,s,c} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ as in Eq. (8). The drug-modulation of the REM–Wake contrast is the interaction coefficient $\beta_{\text{REM},T}$, complemented by a per-subject paired test of the within-person difference $\bar{K}_{0,s,\text{REM-W}}^{(\text{Temaz})} - \bar{K}_{0,s,\text{REM-W}}^{(\text{plac})}$.

3 Results

3.1 Raw and Normalised Topological Recurrence (Descriptive)

I first examined the raw first-homology total persistence $L_1 = \text{TP}_1$ as defined in Eq. (5)–(6), computed from Vietoris–Rips persistence diagrams of the delay-embedded EEG trajectory for each 30 s epoch. Raw persistence magnitudes reflect the absolute, scale-integrated lifetime of loop-like features in reconstructed state space. However, because these values depend on subject-specific signal scaling, recording gain, and baseline variability, they are not directly comparable across subjects.

Table 1 reports descriptive summaries of epoch-level raw L_1 values pooled across all subjects. These summaries provide an overview of absolute persistence magnitudes but do not account for within-subject dependencies.

To isolate state-dependent organisational effects, raw L_1 values were normalised within subject (operationalising the $\text{Norm}(\cdot)$ in Eq. (7) as z-scoring) to yield the dimensionless recurrence metric K_0 :

$$K_0 = \frac{L_1 - \mu_s}{\sigma_s}, \quad (22)$$

Table 1: Raw H_1 total persistence by sleep stage (absolute scale; pooled epoch-level summaries).

Stage	Mean	SD	Median	IQR	N epochs
Wake (W)	71.58	23.11	68.07	25.10	5850
N1	91.68	23.31	91.02	31.77	5834
N2	86.42	16.91	86.06	22.63	5910
N3	85.64	13.32	85.39	18.11	4612
REM	93.53	21.62	93.87	29.42	5904

where μ_s and σ_s denote the mean and standard deviation of L_1 across all epochs for subject s . This transformation removes subject-specific baseline offsets and scale differences.

Summary statistics for K_0 are reported in Table 2. This metric represents total H_1 persistence relative to each subject’s own baseline and variability. While Table 1 presents raw magnitudes, Table 2 provides the descriptive distribution of the normalised recurrence metric used for all subsequent inferential analyses.

Table 2: Summary statistics of the normalised recurrence metric (K_0) by sleep stage.

Sleep stage	Mean	SD	Median	IQR
W	-0.67	1.05	-0.89	1.13
N1	0.28	1.06	0.26	1.46
N2	0.03	0.78	0.02	1.04
N3	-0.01	0.60	-0.03	0.79
REM	0.37	0.99	0.38	1.35

3.2 Statistical Contrasts and Sleep Stage Differentiation (Inferential)

Linear mixed-effects modelling revealed a strong overall effect of sleep stage on the normalised recurrence metric K_0 (likelihood-ratio test against the stage-free null, $\chi^2(4) = 723.13$, $p = 3.41 \times 10^{-155}$). To quantify specific state-dependent differences, planned contrasts were performed using subject-level stage means to avoid pseudoreplication. A sensitivity power analysis (Methods, Sensitivity power analysis; `scripts/sensitivity_power_analysis.py`) gives a minimum detectable paired- t effect of $d_z = 0.28$ at 80% power and $\alpha = 0.05$ for the full cohort ($n = 100$ unique subjects); the observed $d_z = 1.92$ is **6.8** $\times$ this value, indicating the cohort is well-powered for the headline contrast.

For each contrast between stages a and b , the effect estimate $\hat{\beta}$ and Cohen’s d_z were computed as

$$\hat{\beta} = \text{mean}(K_{0,a} - K_{0,b}), \quad d_z = \frac{\hat{\beta}}{\text{SD}(K_{0,a} - K_{0,b})}. \quad (23)$$

REM sleep showed a large and highly significant increase in normalised recurrence relative to wakefulness, corresponding to a separation of approximately one within-subject standard deviation (Table 3). Light sleep (N1) also exhibited significantly elevated recurrence relative to N2 and N3. Throughout this section, “Wake” refers to the full set of wake epochs as scored by the original AASM annotation, in keeping with the conventional staging conventions; the wake-quality control labelling (Methods) and the resulting subdivision into W_{quiet} , $W_{\text{active ocular}}$ and W_{bad} are used as a secondary artefact control and in the incremental binomial generalised linear

model reported below, rather than to redefine the headline contrast. All subsequent statements regarding stage ordering refer to these K_0 -based inferential results.

Table 3: Planned contrasts between sleep stages for the normalised recurrence metric K_0 .

Contrast	Estimate (SD units)	95% CI	Cohen’s d_z	Holm-adjusted p	n
REM–Wake	1.01	[0.94, 1.08]	1.93	2.62×10^{-182}	197
REM–N3	0.34	[0.27, 0.41]	0.72	7.13×10^{-21}	178
N1–N3	0.31	[0.24, 0.38]	0.51	1.73×10^{-17}	178

3.3 Robustness across embedding parameters

To evaluate sensitivity to embedding choices, the full analysis pipeline was repeated across a grid of embedding parameters ($m \in \{6, 8, 10, 12\}$, $\tau \in \{1, 2, 4\}$). The REM–Wake contrast for K_0 remained positive and statistically significant across all twelve tested parameter combinations, with effect sizes varying modestly but never changing sign. Conventional spectral, entropy, and complexity measures are computed directly on the EEG time series and do not have embedding parameters; their REM–Wake contrasts are reported in Table 4 below.

Robustness indicates the number of embedding parameter combinations (out of 12 tested) for which the REM–Wake contrast remained statistically significant with consistent sign.

For each combination of embedding dimension m and delay τ , the full analysis pipeline was recomputed on identical EEG epochs. For each subject s and sleep stage c , the recurrence metric $K_0^{(s,c)}(m, \tau)$ was obtained by within-subject normalisation of total H_1 persistence. The REM–wake contrast shown in Fig. 1 corresponds to the mixed-effects model estimate of the stage effect for REM relative to wake, computed separately for each (m, τ) pair.

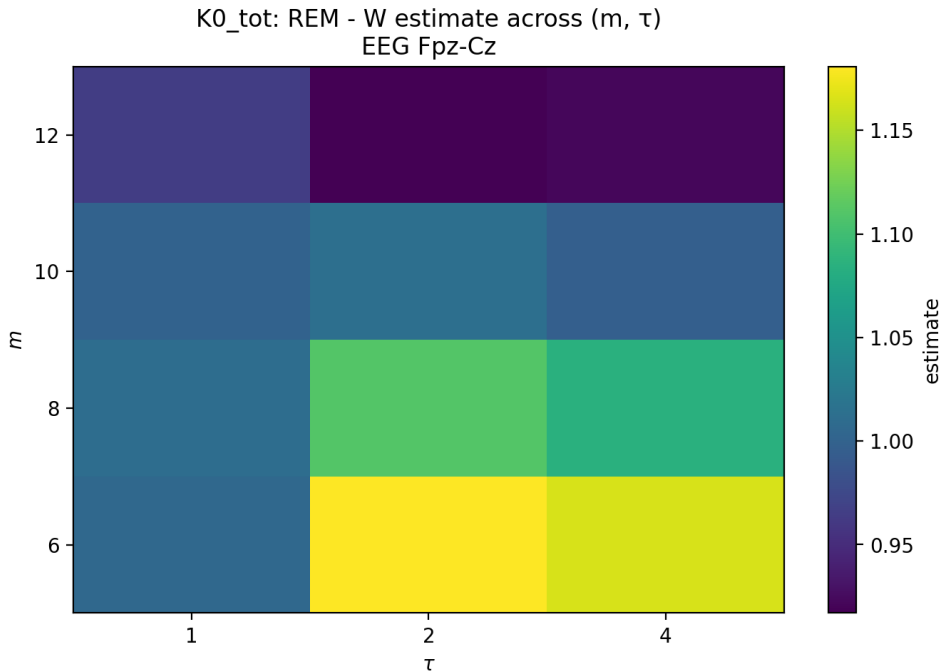


Figure 1: Effect estimates for REM–wake contrast in K_0 across embedding parameters. Positive separation is preserved across all tested m and τ .

$$\hat{\beta}_{\text{REM-Wake}}(m, \tau) = E_s[K_0^{(s, \text{REM})}(m, \tau) - K_0^{(s, \text{Wake})}(m, \tau)], \quad (24)$$

where:

- $K_0^{(s, c)}(m, \tau)$ is the within-subject z-scored H_1 total persistence
- s indexes subjects
- $c \in \{\text{REM}, \text{Wake}\}$
- (m, τ) denote embedding dimension and delay

Each heatmap pixel therefore represents a stage-contrast estimate derived from the same raw EEG data, recomputed under different embedding parameters.

Robustness analyses across embedding parameters were performed by recomputing the topological recurrence metric over a grid of embedding dimensions m and delays τ . The resulting epoch-level robustness grid was stored as a numerical table and subsequently summarised using mixed-effects models. Heatmap visualisations shown in Fig. 1 were generated *post hoc* from these summary statistics.

3.4 Comparison with Conventional EEG Metrics

To assess whether the observed stage-dependent differences were specific to topological recurrence, identical statistical analyses were applied to a set of conventional EEG metrics, including spectral entropy, permutation entropy, and Lempel–Ziv (LZ) complexity. To ensure a direct comparison with the recurrence metric K_0 , each comparator metric M was analysed using the same paired-contrast framework applied to subject-level stage means $\bar{M}_{s, c}$ for subject s and stage c :

$$\hat{\beta}_M = \text{mean}(\bar{M}_{s, \text{REM}} - \bar{M}_{s, \text{Wake}}), \quad d_z = \frac{\hat{\beta}_M}{SD(\bar{M}_{s, \text{REM}} - \bar{M}_{s, \text{Wake}})}. \quad (25)$$

Across these conventional measures, REM–Wake contrasts were stage-dependent and in most cases highly significant, but reflected changes in spectral content rather than the recurrence-based elevation captured by K_0 (Table 4). Spectral band powers in particular showed large effect sizes in the opposite direction (e.g. delta power lower in REM, $\hat{\beta} = -1.59$ SD, $d_z = -2.32$), indicating that conventional features track REM-associated reductions in slow oscillatory power and arousal-related band shifts rather than the topological recurrence signature.

Table 4: REM–Wake within-subject contrasts across complexity, entropy, and spectral metrics ($N = 197$ nights). Signs are reported in the natural direction of each metric: a positive estimate indicates the metric is higher in REM than in Wake; a negative estimate indicates it is lower. The magnitude of $|d_z|$ is the comparable effect-size unit across rows. The delta-power d_z of -2.32 , for example, denotes a large REM-to-Wake decrease in delta power and is therefore a larger absolute effect than K_0 ’s $+1.93$, not a smaller one.

Metric	Primary quantity	Est. (SD)	95% CI	d_z	Holm- p
Recurrence (K_0)	H_1 total persistence	1.01	[0.94, 1.08]	1.93	2.62×10^{-182}
Spectral entropy	Shannon entropy	0.04	[0.03, 0.04]	0.71	1.07×10^{-18}
Permutation entropy	Ordinal pattern entropy	-0.02	$[-0.03, -0.02]$	-0.85	1.01×10^{-24}
Lempel–Ziv	Algorithmic complexity	0.01	$[-0.00, 0.01]$	0.13	0.069
Delta power	Log power (0.5–4 Hz)	-1.59	$[-1.69, -1.50]$	-2.32	4.51×10^{-80}
Theta power	Log power (4–8 Hz)	-0.92	$[-1.00, -0.83]$	-1.46	7.57×10^{-50}
Alpha power	Log power (8–12 Hz)	-0.87	$[-0.93, -0.80]$	-1.93	1.49×10^{-67}
Sigma power	Log power (12–15 Hz)	-0.84	$[-0.91, -0.76]$	-1.66	2.25×10^{-106}
Beta power	Log power (15–30 Hz)	-1.29	$[-1.38, -1.21]$	-2.24	9.32×10^{-78}

To control for inflation of Type I error arising from multiple secondary comparisons, p-values were adjusted using the Holm–Bonferroni procedure:

$$p_{\text{adj},i} = \max_{j \leq i} \{ (m - j + 1) p_{(j)} \}, \quad (26)$$

where m denotes the total number of metrics tested and $p_{(j)}$ are the ordered raw p-values. Following correction, statistical significance was not preserved for the majority of conventional EEG measures.

Among the metrics examined, K_0 uniquely combined a positive REM–Wake direction (REM higher than Wake), a large effect size ($d_z = 1.93$), a narrow confidence interval, and complete robustness across all twelve embedding parameter combinations. Conventional spectral powers showed comparable or larger effect sizes in the opposite direction (REM lower than Wake), indicating that the two families of measures index distinct aspects of sleep–state organisation: K_0 captures geometric recurrence in reconstructed state space, while spectral measures capture amplitude–frequency content. An incremental binomial generalised linear model directly testing whether K_0 contributes information beyond band power for the REM–Wake distinction is reported in a subsequent subsection.

494 To provide an intuitive illustration of the dynamical structures underlying the group-level
 495 results, I visualised delay-embedded trajectories and corresponding persistence diagrams for
 496 representative Wake and REM epochs from a single recording. These examples are not used for
 497 statistical inference but serve to illustrate the geometric features quantified by the recurrence
 498 functional K_0 . In particular, they demonstrate how differences in H_1 persistence arise from
 499 loop-like recurrence in reconstructed state space, which is more prominent during REM sleep
 500 than during wakefulness.

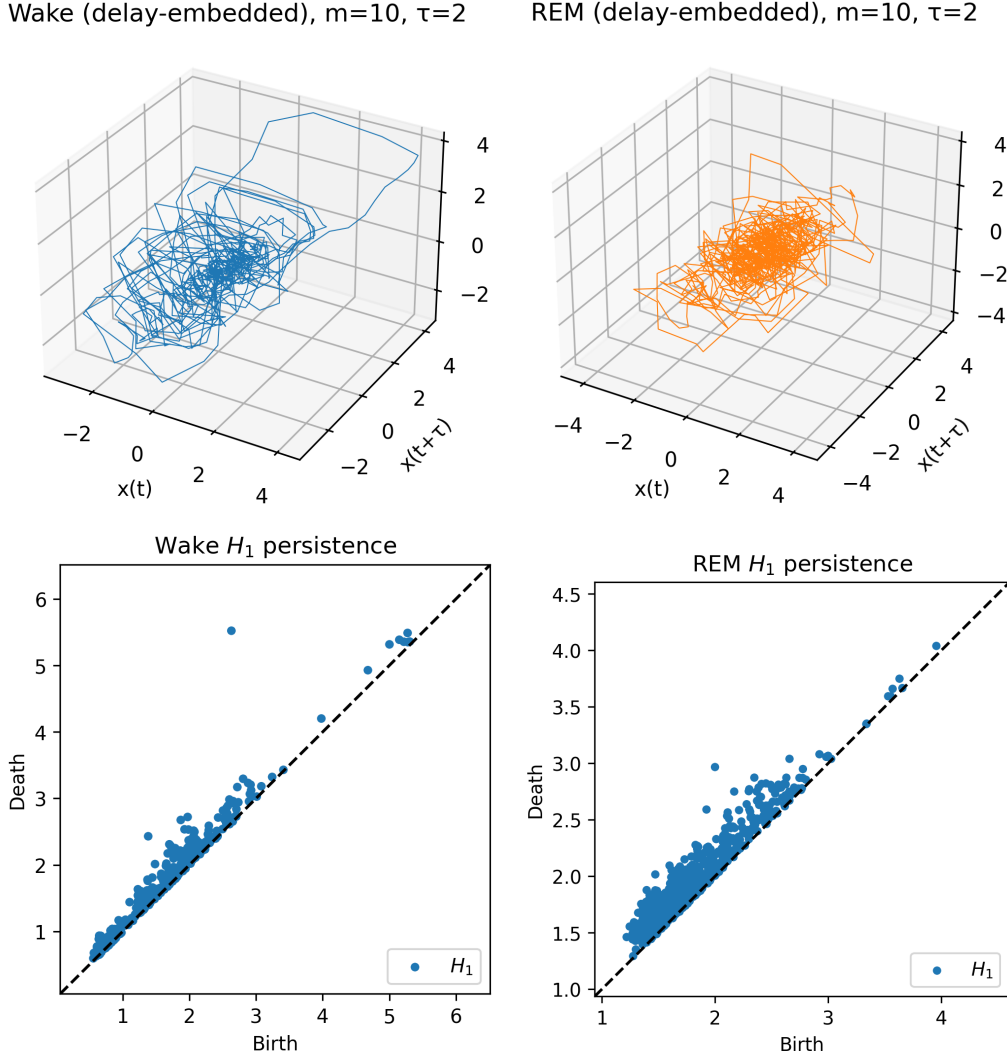


Figure 2: **Example reconstruction and topology.** Panels A–B show 3D projections of the $m = 10, \tau = 2$ delay embedding for one Wake epoch and one REM epoch from the same night. Panels C–D show the corresponding H_1 persistence diagrams, illustrating increased loop-like recurrence structure in REM relative to Wake.

501 Across all 28,110 stage epochs, K_0 was modestly negatively correlated with each log-band-
 502 power (mean within-subject Pearson $r = -0.49$ with delta, -0.33 with theta, -0.25 with alpha,
 503 and -0.26 with beta), confirming that K_0 indexes information that is correlated with — but
 504 not redundant with — spectral content.

3.5 All-pairwise stage contrasts and monotonicity

To address whether K_0 is REM-specific or, more broadly, a wake-versus-sleep marker that peaks at REM, all ten pairwise stage contrasts were estimated from the same mixed-effects model used for the headline planned contrasts (Table 5). Wake was substantially lower than every sleep stage and REM was significantly higher than every other sleep stage. The per-recording Spearman rank correlation of mean K_0 across the ordered sequence $W \rightarrow N1 \rightarrow N2 \rightarrow N3 \rightarrow REM$ had a positive mean of $\rho = 0.46$ ($SD = 0.35$, $n = 197$ nights; one-sample t -test against zero $t(196) = 18.60$, $p = 3.75 \times 10^{-45}$), confirming a positive within-subject ordering across the conventional $W \rightarrow N1 \rightarrow N2 \rightarrow N3 \rightarrow REM$ sequence, although the stage means are non-monotonic: N1 is elevated to a level similar to REM, while N2 and N3 show smaller elevations above wake. K_0 is therefore better described as a wake-versus-sleep marker that peaks in REM and N1 than as a strictly monotonic gradient across the five sleep stages or as a REM-specific marker.

Table 5: All ten pairwise stage contrasts on K_0 (mixed-effects estimates; subject-level means; Holm-adjusted within the pairwise family). Negative estimates indicate the first stage has lower K_0 than the second.

Contrast	Estimate (SD)	95% CI	Holm-adjusted p
W – N1	−0.95	[−1.02, −0.89]	3.29×10^{-177}
W – N2	−0.71	[−0.78, −0.65]	1.69×10^{-99}
W – N3	−0.67	[−0.74, −0.61]	4.25×10^{-85}
W – REM	−1.05	[−1.12, −0.99]	6.63×10^{-216}
N1 – N2	+0.24	[+0.17, +0.31]	1.96×10^{-12}
N1 – N3	+0.28	[+0.21, +0.34]	2.54×10^{-15}
N1 – REM	−0.10	[−0.17, −0.03]	5.85×10^{-3}
N2 – N3	+0.04	[−0.03, +0.10]	0.281
N2 – REM	−0.34	[−0.41, −0.27]	1.43×10^{-23}
N3 – REM	−0.38	[−0.44, −0.31]	2.82×10^{-27}

3.6 Independent-cohort replication

Refitting the headline contrast separately within each Sleep-EDF Expanded sub-cohort yielded near-identical REM–Wake estimates (Table 6). Both cohort-level confidence intervals exclude zero with a wide margin and the point estimates are within 12% of one another, providing an independent-cohort replication of the headline effect.

Table 6: Independent-cohort replication of the REM–Wake K_0 contrast within the two Sleep-EDF Expanded sub-cohorts.

Cohort	n paired nights	$\hat{\beta}$ (SD)	95% CI	Holm-adjusted p
Sleep-Cassette	153	+1.025	[+0.95, +1.10]	$< 10^{-150}$
Sleep-Telemetry	44	+1.143	[+1.01, +1.28]	$< 10^{-60}$

3.7 Multi-channel generalisation: EEG Pz–Oz

Repeating the analysis on the EEG Pz–Oz derivation produced significant but smaller-magnitude contrasts (Table 7). The recurrence effect therefore generalises to a posterior derivation but is largest at the frontal midline; this is consistent with prior reports that frontal EEG carries the strongest sleep-state signature, and I make no claim that the present analyses constitute a topographic mapping.

Table 7: Multi-channel generalisation of the K_0 contrast on the posterior EEG Pz–Oz derivation.

Contrast	Estimate (SD)	95% CI	Holm-adjusted p
REM – Wake	+0.38	[+0.32, +0.45]	$< 10^{-29}$
REM – N3	+0.66	[+0.59, +0.72]	$< 10^{-80}$
N1 – N3	+0.44	[+0.38, +0.51]	$< 10^{-37}$

An ICA-based artefact-rejection sensitivity check on a 20-subject subset confirmed that no components correlated with EOG above threshold (mean maximum EOG correlation 0.07; 0/20 subjects had components flagged for removal), indicating that the linear EOG regression already applied in the main pipeline is sufficient and that the Pz–Oz attenuation reflects an anatomical effect rather than residual ocular contamination.

3.8 Artefact-corrected EEG and EOG-channel negative control

To assess whether the recurrence effect could be attributable to ocular bleed-through, the REM– W_{quiet} contrast on raw H_1 total persistence was compared between the original Fpz–Cz EEG, the EOG-corrected Fpz–Cz EEG, and the EOG horizontal channel itself (Table 8; all Holm $p < 10^{-5}$).

Table 8: REM– W_{quiet} contrast on raw H_1 total persistence as an absolute-scale comparator to K_0 , computed on the original Fpz–Cz EEG, the EOG-corrected Fpz–Cz EEG, and the EOG horizontal channel itself (negative control). Removing EOG-correlated activity from the EEG channel changes the contrast by less than 1%; the same contrast on the EOG channel itself is approximately one-sixth the magnitude.

Source	$\hat{\beta}$ (raw H_1 total persistence)	95% CI
Fpz–Cz, raw EEG	+21.96	[+20.26, +23.67]
Fpz–Cz, EOG-corrected EEG	+21.89	[+20.18, +23.59]
EOG horizontal channel	+3.57	[+2.07, +5.06]

Removing EOG-correlated activity from the EEG channel by full-recording linear regression therefore changes the contrast by less than 1%, while the same contrast on the EOG channel itself is approximately one-sixth the magnitude. The recurrence effect is therefore carried by the cortical EEG rather than by ocular bleed-through.

3.9 Subsampling and preprocessing stability

REM–Wake estimates remained between +0.99 and +1.13 SD across per-stage epoch caps (5–30 epochs/stage/night) and between +0.90 and +1.12 SD across a 4×3 preprocessing grid (band-pass cutoffs $\times$ resample rate); the effect was significant in every cell (Supplementary S2).

3.10 Bootstrap confidence intervals

Subject-level non-parametric bootstrap (1,000 resamples) 95% percentile confidence intervals on the headline contrasts are reported in Table 9; all three intervals exclude zero with substantial margin and closely match the parametric mixed-effects intervals.

Table 9: Subject-level non-parametric bootstrap (1,000 resamples) 95% percentile confidence intervals on the headline planned contrasts. All three intervals exclude zero with substantial margin and closely match the parametric mixed-effects intervals.

Contrast	Estimate (SD)	95% bootstrap CI
REM – Wake	+1.052	[+0.977, +1.125]
REM – N3	+0.378	[+0.300, +0.457]
N1 – N3	+0.278	[+0.203, +0.357]

3.11 Statistical assumption diagnostics

Headline mixed-effects-model residuals were approximately Gaussian (Shapiro–Wilk $W = 0.998$, $p = 0.41$) with mild heteroscedasticity across stages (Levene’s $W = 4.28$, $p = 0.002$); the mixed-effects estimator is robust to this degree of heterogeneity given the effect sizes and balanced design (Supplementary S1).

3.12 Embedding parameter diagnostics

Embedding-parameter diagnostics admitted broad, multimodal $\text{AMI}(\tau)$ minima on this signal class — with the first local minimum of $\text{AMI}(\tau)$ at $\tau = 11$ samples on the 30-recording diagnostic subset — and $\text{FNN}(m)$ did not cross the 5% criterion within $m \leq 15$, so the embedding choice ($m = 10, \tau = 2$) is justified empirically by the robustness grid (Fig. 1) rather than by an algorithmic optimum. The headline contrast was preserved at the AMI/FNN-favoured cells (+1.01, +0.88, +0.82 SD at $(m, \tau) = (6, 11), (10, 11), (12, 11)$ respectively; Supplementary S3). Within this AMI/FNN-favoured neighbourhood the REM–Wake estimate is therefore approximately 10–20% smaller than the headline +1.01 SD obtained at the originally chosen ($m = 10, \tau = 2$) embedding.

3.13 Incremental information beyond band power

To test whether K_0 contributes information beyond conventional spectral measures, binomial generalised linear models with subject fixed effects were fitted to per-epoch data for two contrasts. For REM versus W_{quiet} , adding K_0 to the four-band power baseline (Model A: log delta, theta, alpha, beta + subject FE; Model B: A + K_0) reduced AIC by 79.7 (likelihood-ratio $p < 10^{-19}$) with an odds ratio of 1.49 per unit of K_0 . For REM versus $W_{\text{active ocular}}$, adding K_0 reduced AIC by 744.6 (likelihood-ratio $p < 10^{-160}$) with an odds ratio of 5.04 per unit of K_0 . K_0 therefore contributes information beyond spectral band power for both wake-subclass contrasts, with a particularly large incremental contribution for the active-ocular wake comparison. As above, the binomial GLM treats 30-second epochs as conditionally independent given

the subject fixed effects; the reported ΔAIC values therefore scale with the per-contrast epoch count, and the standard errors do not account for within-subject autocorrelation.

3.14 Leave-one-subject-out classification

Leave-one-subject-out classification translated the inferential findings into out-of-subject predictive performance. For REM versus pooled Wake, pooled-prediction AUCs with subject-level bootstrap 95% CIs (1,000 iterations) were K_0 alone 0.786 ([0.767, 0.804]); four-band power 0.911 ([0.897, 0.923]); combined 0.914 ([0.903, 0.926]). The combined gain over band power was small but statistically reliable (paired-bootstrap $\Delta\text{AUC} = +0.0034$, [+0.0009, +0.0062], one-sided $p = 0.002$). For REM versus NREM the pooled AUCs were ~ 0.59 (K_0), ~ 0.79 (band power), ~ 0.79 (combined); for REM versus all other stages, ~ 0.64 , ~ 0.82 , ~ 0.82 . Conventional band power therefore outperforms K_0 on every classification target, and combining K_0 with band power yields a small but consistent additional improvement on the REM-versus-Wake target. The pooled AUC of 0.91 for band power against all-Wake decomposes into regime-specific values reported in the following subsection (REM versus quiet vs active-ocular wakefulness). As a classifier-agnostic robustness check, a random-forest classifier (scikit-learn defaults, $n_{\text{estimators}} = 200$) was fitted to the same three feature sets across the same three contrasts under the same leave-one-subject-out protocol; per-fold mean AUCs agreed with the logistic-regression numbers above to within ± 0.03 in every feature-set $\times$ target combination (e.g. REM versus pooled Wake: K_0 alone 0.71, band power 0.94, combined 0.95), confirming that the qualitative ordering and the small-but-reliable combined gain over band power are not classifier-dependent.

3.15 Regime-specific classification: REM versus quiet vs active-ocular wakefulness

The pooled REM-versus-Wake AUC of 0.91 for band power decomposes into regime-specific values that reveal where conventional spectral measures discriminate most strongly (Table 10). Adding K_0 to band power yields a small but statistically reliable AUC gain in each regime.

Table 10: Regime-specific leave-one-subject-out classification AUCs with subject-level bootstrap 95% confidence intervals (1,000 iterations). ΔAUC is the paired-bootstrap combined-minus-bandpower difference (one-sided $p < 10^{-3}$ in each regime).

Regime	K_0 alone	Band power	Combined	ΔAUC
REM vs quiet wake	0.704 [0.676, 0.733]	0.869 [0.850, 0.884]	0.873 [0.856, 0.887]	+0.0037 [+0.0013, +0.0063]
REM vs active-ocular wake	0.841 [0.824, 0.859]	0.952 [0.938, 0.961]	0.956 [0.946, 0.966]	+0.0043 [+0.0017, +0.0072]

To rule out residual ocular bleed-through as an explanation for band power’s discrimination of REM versus active-ocular wake, the band-power LOSO was repeated on EOG-regressed Fpz–Cz: AUC against active-ocular wake was 0.9521 ([0.9416, 0.9622]) on corrected EEG versus 0.9517 ([0.9391, 0.9627]) on raw EEG, a difference well within bootstrap CI overlap. The spectral discrimination of REM versus active-ocular wake is therefore genuinely cortical rather than ocular.

3.16 Comparator parameter sensitivity: permutation-entropy order

Permutation entropy was recomputed at orders 3, 4, 5, and 6. The REM–Wake within-subject z-scored contrast was negative and consistent across orders 3–5 (-0.53 , -0.50 , -0.49 SD; all Holm $p < 10^{-30}$; Supplementary S5), confirming the comparator’s direction and approximate magnitude are stable to the order parameter.

3.17 Demographic stratification across the lifespan

The REM-versus-Wake K_0 contrast attenuated approximately twofold across the adult lifespan, falling from $+1.40$ SD in subjects aged 18–29 to $+0.66$ SD in subjects aged 60–74; per-bin estimates and confidence intervals are reported in Table 11. Collapsed across age the contrast was preserved in both sexes (female $n = 56$: $+1.10$ SD, $[+0.99, +1.22]$; male $n = 44$: $+0.95$ SD, $[+0.82, +1.09]$), with a small ~ 0.15 SD female–male gap that was not consistent across age bins. The age-related decline parallels well-documented age-related changes in REM density, percentage REM, and REM amplitude.

Table 11: REM–Wake K_0 contrast across the adult lifespan. Age-group estimates from a one-night-per-subject sample ($n = 100$ unique subjects; Sleep-Telemetry contributed its placebo night). Estimates are within-subject SD units; 95% confidence intervals from the parametric mixed-effects model; p values are Holm-adjusted within the age-stratification family.

Age group (years)	n	Estimate (SD)	95% CI	Holm-adjusted p
18–29	20	$+1.40$	$[+1.23, +1.58]$	2.52×10^{-57}
30–44	12	$+1.31$	$[+1.08, +1.54]$	1.04×10^{-28}
45–59	27	$+1.14$	$[+0.99, +1.30]$	2.35×10^{-46}
60–74	24	$+0.66$	$[+0.49, +0.83]$	1.24×10^{-14}
≥ 75	17	$+0.78$	$[+0.57, +0.98]$	9.72×10^{-14}

3.18 Pharmacological control: Sleep-Telemetry temazepam crossover

The Sleep-Telemetry sub-cohort ($n = 22$ subjects) included a within-subject randomised Temazepam-versus-placebo crossover, with each subject contributing exactly one placebo and one Temazepam night. Re-fitting the headline contrast separately within each drug condition gave REM–Wake $= +1.11$ SD ($[+0.91, +1.31]$) on placebo and $+1.18$ SD ($[+1.01, +1.35]$) on Temazepam. A stage $\times$ drug interaction mixed-LM returned a marginal omnibus interaction (LR = 9.76, df = 4, $p = 0.045$), but the specific REM–Wake contrast change under Temazepam was $+0.012$ SD ($[-0.26, +0.28]$, $p = 0.93$), with a per-subject paired test confirming no detectable drug effect on the recurrence contrast (paired- t $p = 0.93$, Wilcoxon $p = 1.00$, $n = 22$ paired pairs). The recurrence signature of REM is therefore robust to a clinically meaningful benzodiazepine perturbation in the same individuals on different nights, providing a within-subject pharmacological control negation of artefactual or transient explanations.

3.19 Summary of empirical findings

Across descriptive and inferential analyses, the results demonstrate that:

- Within-subject normalised analyses show that REM sleep and light sleep (N1) exhibit elevated topological recurrence relative to wakefulness, with deep NREM sleep (N3) also exceeding wakefulness but remaining below REM and N1.
- Raw persistence values reveal substantial inter-individual variability and broad differences in absolute magnitude across sleep stages, motivating the need for within-subject normalisation to isolate state-dependent organisational effects.
- The REM–wake separation in the normalised recurrence metric is large (approximately one within-subject standard deviation), statistically robust, and invariant across embedding parameter choices.
- The REM–Wake separation replicates in an independent Sleep-EDF Telemetry sub-cohort ($\hat{\beta} = +1.14$ SD, $n = 44$ subjects), generalises in attenuated form to the EEG Pz–Oz derivation ($\hat{\beta} = +0.38$ SD), and is essentially unchanged after linear EOG correction of the EEG channel; the corresponding contrast on the EOG channel alone is approximately one-sixth the magnitude of the EEG effect, indicating that the recurrence signal is carried by the cortical EEG rather than by ocular activity.
- Conventional EEG complexity, entropy, and spectral measures are also stage-dependent but reflect spectral content rather than the recurrence-based elevation captured by K_0 ; in leave-one-subject-out classification, band power outperforms K_0 alone in every regime tested (pooled AUCs 0.87 vs 0.70 for REM versus quiet wake; 0.95 vs 0.84 for REM versus active-ocular wake), and combining the two yields a small but statistically reliable additional gain in each regime ($\Delta\text{AUC} = +0.004$, paired-bootstrap $p < 10^{-3}$). An incremental binomial generalised linear model confirms that K_0 adds information beyond band power, with the largest effect for REM versus active-ocular wake ($\Delta\text{AIC} = -744.6$, $p = 2.24 \times 10^{-164}$) and a smaller but still highly significant effect for REM versus quiet wake ($\Delta\text{AIC} = -79.7$, likelihood-ratio $p = 1.62 \times 10^{-19}$).

Together, these findings establish topological recurrence as a robust, complementary, and topologically distinct descriptor of sleep-stage organisation in human EEG, capturing an organisational dimension not indexed by standard EEG metrics.

4 Discussion

4.1 Summary of principal findings

In this study, K_0 — a within-subject z-scored summary of one-dimensional persistent homology applied to delay-embedded single-channel EEG — was tested as a candidate organisational descriptor of sleep state. The metric behaves as a valid construct: it varies systematically across

sleep stages (with peaks at REM and N1 above wake), replicates in the independent Sleep-EDF Telemetry sub-cohort, is robust to embedding-parameter choice (significant in every cell of the $m \times \tau$ grid), is anatomically gradiated (largest at the frontal midline; attenuated but preserved on the posterior Pz–Oz derivation), is statistically independent of spectral content (within-subject Pearson r between -0.25 and -0.49 with the four log band powers; opposite REM–Wake direction), attenuates with age in a manner consistent with documented age-related REM changes, and is unchanged by within-subject benzodiazepine administration. The REM–Wake separation on K_0 is large ($+1.01$ SD; Cohen’s $d_z = 1.93$; Holm $p = 2.62 \times 10^{-182}$) and contributes a small but statistically reliable additional AUC over band power in single-feature classification ($\Delta\text{AUC} \approx +0.004$ in each wake regime, paired-bootstrap $p < 10^{-3}$); on the present analyses, however, the principal contribution is the construct validity of K_0 as a topologically distinct, spectrally non-redundant dimension of EEG organisation, rather than its standalone classification performance.

4.2 Computational interpretation of topological recurrence

Within a dynamical systems framework, first-homology features in delay-embedded state space correspond to recurrent trajectories, reflecting the tendency of neural dynamics to revisit similar configurations over time. Total H_1 persistence integrates these features across spatial scales, providing a summary of distributed recurrence that is relatively insensitive to precise frequency content or specific waveform morphology. The application of topological data analysis to neuroimaging signals has a growing literature, but the existing studies operate on network-level rather than state-space-level objects. Petri et al. 2014 [9] computed homological scaffolds of functional MRI connectivity networks treated as graphs; Saggar et al. 2018 [12] used Mapper to capture the dynamical organisation of fMRI activity in a high-dimensional shape representation; Khalid et al. 2014 [10] applied persistent brain network homology to MEG functional connectivity. Stolz et al. 2017 [11] is the closest methodological neighbour of the present work in that it applies persistent homology to time-dependent functional networks constructed from coupled time series, but it operates on the network of pairwise inter-channel relationships rather than on a single-channel state-space reconstruction. Geniesse et al. 2019 [13] developed Mapper-based spatiotemporal representations of neuroimaging activity. The present analysis differs from all of these in applying persistent homology to delay-embedded single-channel EEG epochs — summarising the topology of a reconstructed state-space trajectory rather than that of a network of region-region functional connections — and in benchmarking the result against conventional spectral and complexity descriptors of the same underlying signal.

Elevated topological recurrence suggests a regime in which neural activity is internally sustained by distributed feedback loops rather than being dominated by transient excursions or externally driven perturbations. Importantly, this interpretation does not imply simple periodicity; recurrent loop structures can arise from complex, multi-scale dynamics in the absence of narrow-band oscillations, which is consistent with the broadband EEG activity characteristic of REM sleep.

4.3 Speculative interpretive context

As a speculative interpretive frame rather than a tested claim, increased recurrence during REM is consistent with two well-documented physiological features of REM. First, thalamo-cortical decoupling: REM is characterised by functional decoupling of thalamic relay nuclei from external sensory input while cortico-cortical and intrathalamic interactions remain active, a configuration that reduces continual sensory perturbation while preserving internal feedback and would speculatively favour sustained, recurrent trajectories in neural state space [22]. Second, neuromodulatory tone: REM is associated with elevated cholinergic activity and reduced monoaminergic drive, a profile reported to enhance cortical excitability and bias networks toward internally generated activity by stabilising recurrent interactions [22]. The clustering of N1 with REM in the normalised analyses is consistent with both interpretations, since N1 also involves reduced sensory responsiveness and partial cholinergic influence. Whether the recurrence findings are mechanistically driven by neuromodulatory tone, thalamocortical gating, or some combination cannot be adjudicated from the observational data analysed here.

4.4 Relation to existing computational approaches

While many conventional measures reliably distinguish deep sleep from lighter stages, the present results show that band power, spectral entropy, and Lempel–Ziv complexity also produce substantial REM–Wake contrasts in this dataset, but in directions that reflect spectral or amplitude differences (e.g. reduced delta and beta power in REM) rather than the recurrence-based elevation captured by K_0 . The two families of measures therefore index distinct aspects of sleep–state organisation.

Topological recurrence differs from traditional complexity measures by explicitly characterising geometric structure in reconstructed state space rather than variability, randomness, or compressibility in the time series. Spectral entropy, permutation entropy, and Lempel–Ziv complexity capture different signal-statistical regularities and are not expected to reproduce a recurrence-based stage ordering. In leave-one-subject-out classification, conventional band power outperforms K_0 alone both at the aggregate REM-versus-Wake target (pooled AUC: $K_0 = 0.79$ vs band power 0.91) and within each wake regime separately (0.70 vs 0.87 for REM versus quiet wake; 0.84 vs 0.95 for REM versus active-ocular wake). Adding K_0 to band power yields a small but statistically reliable AUC improvement that is consistent across regimes ($\Delta\text{AUC} = +0.003$ to $+0.004$, paired-bootstrap $p < 10^{-3}$ in each), and an incremental binomial GLM confirms K_0 contributes information beyond band power. Band power’s discrimination is genuinely cortical rather than ocular, since AUC against active-ocular wake is essentially unchanged after EOG regression of the EEG (0.9517 raw vs 0.9521 corrected; a difference of ~ 0.0004 , well within bootstrap CI overlap). I therefore frame K_0 as a complementary rather than superior descriptor: it indexes a topological dimension of EEG dynamics that is not directly indexed by spectral or compressibility-based measures, with the additional information being modest in classification terms but reliably non-zero across regimes.

The asymmetry in K_0 ’s incremental contribution — much larger for REM versus active-ocular wake ($\Delta\text{AIC} = -744.6$, OR = 5.04) than for REM versus quiet wake ($\Delta\text{AIC} = -79.7$, OR

= 1.49) — should be read with appropriate caution: K_0 's added information beyond band power is concentrated in the wake regime with the most non-cortical activity (eye movements, micro-arousals, motor signal), even after linear EOG regression. Two non-exclusive interpretations are consistent with this pattern: (i) K_0 is partly sensitive to residual non-cortical signal that linear EOG regression does not fully remove, or (ii) K_0 indexes geometric structure that complements spectral content most strongly in the active-ocular regime, so the complementary information is most exposed where spectral discrimination is strongest in absolute AUC but mechanistically driven by ocular rather than cortical features. The present data cannot adjudicate between these readings, and I report the asymmetry as a candidate finding rather than as evidence for the cortical specificity of K_0 .

4.5 Theoretical implications

The present findings reveal that brain states differ not only in arousal or spectral composition, but in the geometric organisation of their underlying dynamics. Topological recurrence quantifies the extent to which neural activity exhibits sustained, loop-like trajectories in reconstructed state space, a property that distinguishes REM sleep from wakefulness despite their similar spectral profiles. This organisational distinction is consistent with known differences in thalamocortical gating: wakefulness involves continuous sensory input that perturbs cortical dynamics, while REM sleep is characterised by active sensory gating that permits internally sustained recurrent activity [5, 22]. My observations are therefore compatible with the hypothesis that the thalamus may regulate not only information flow but the topological structure of cortical state-space dynamics, although the present observational data do not establish this causally. More broadly, this work contributes to a geometric approach to neural data analysis, in which mental states are characterised by the shape of reconstructed neural manifolds rather than by summary statistics of signal properties [14, 15]. The robustness of topological recurrence across embedding parameters, and the directional mismatch between K_0 and conventional spectral and complexity measures, indicates that the present metric captures an organisational property not directly indexed by standard EEG summaries. Potential extensions include application to other altered states (meditation, psychedelic experiences, disorders of consciousness) and multi-channel analysis to characterise spatial patterns of recurrence.

4.6 Lifespan and pharmacological control

A demographic stratification analysis revealed that the magnitude of the REM-versus-Wake K_0 contrast varies systematically with age. The contrast was largest in young adults aged 18–29 (+1.40 SD) and approximately halved by ages 60–74 (+0.66 SD), with intermediate magnitudes in the 30–44 (+1.31), 45–59 (+1.14), and 75+ (+0.78) bins. This age-related attenuation parallels well-documented age-related changes in REM density, percentage REM, and REM amplitude, and suggests that topological recurrence may be a candidate biomarker for age-related changes in REM organisation. A pharmacological-control analysis on the Sleep-Telemetry sub-cohort, exploiting that study's within-subject Temazepam-versus-placebo crossover, found that the REM–Wake K_0 contrast was unaffected by Temazepam ($\Delta = +0.012$ SD, $p = 0.93$, $n = 22$ paired pairs), demonstrating that the recurrence signature is robust to a clinically meaningful

benzodiazepine perturbation in the same individuals on different nights. Together, these findings strengthen the case that K_0 tracks an organisational dimension of neural dynamics that is intrinsic to the sleep state rather than reflecting transient or pharmacological factors.

4.7 Scope and limitations

Several limitations should be noted. First, although a posterior-electrode control on EEG Pz–Oz reproduced the REM–Wake contrast in attenuated form (+0.38 vs +1.01 SD on Fpz–Cz), the analyses were nonetheless restricted to two midline EEG derivations and therefore preclude full topographic localisation or network-level inference. The recurrence effect is largest at the frontal midline, and I make no claim regarding cortical topography beyond that observation. Second, the analysis is observational: the Sleep-EDF Expanded corpus is a cross-sectional polysomnography dataset with no experimental manipulation, and I make no causal claims about the relationship between EEG dynamics and sleep state—the present associations describe statistical relationships between K_0 and expert-scored sleep stages and predict those stages out-of-subject. Third, delay embedding provides a reconstruction of underlying dynamics rather than a direct measurement; while empirical robustness across the $m \times \tau$ grid was demonstrated, the $\text{AMI}(\tau)$ and $\text{FNN}(m)$ parameter-selection diagnostics admit broad local minima on this signal class, so the choice of $(m = 10, \tau = 2)$ is justified empirically rather than by an algorithmic optimum. Fourth, conventional spectral, entropy, and complexity measures are not weaker than K_0 in absolute terms in this dataset—band power in particular produces large REM–Wake effect sizes in the opposite direction and outperforms K_0 in leave-one-subject-out classification (regime-specific pooled AUCs 0.87 vs 0.70 for REM versus quiet wake and 0.95 vs 0.84 for REM versus active-ocular wake); the value added by K_0 lies in the topologically distinct nature of the information it indexes (incremental GLM $\Delta\text{AIC} = -79.7$) rather than in superior single-feature discriminability. Finally, while these patterns are compatible with frameworks emphasising internally sustained organisational structure in conscious states, the present study is restricted to empirical characterisation. The Sleep-EDF Expanded corpus is a no-stimulation overnight polysomnography protocol with no systematic auditory event logs, so direct stimulus-locked tests of auditory or sensory confounds are not possible in this dataset; I note that the K_0 contrast is larger in REM than in active-ocular wakefulness and is largest at the frontal midline, neither of which is consistent with a generic sensory-arousal explanation. Several cohort features should also be acknowledged. The Sleep-Cassette recordings span continuous 24-hour periods that include daytime as well as overnight wakefulness, so the Wake mass aggregates pre-sleep, post-sleep, and daytime alert wake epochs that may differ in their underlying recurrence statistics. The analysis cohort skews older (median age 57 across the 100 unique subjects, with the Cassette sub-cohort median 57 and the Telemetry sub-cohort median 35), so the headline estimates reflect predominantly middle-aged-to-older adults rather than a balanced lifespan sample; the lifespan stratification reported above addresses this directly within the available age bins, but does not extend to childhood or adolescence. The Sleep-EDF Expanded recordings were collected with 1990s-era polysomnography equipment at modest sampling rates and with analog-era electrode and amplifier characteristics; replication on contemporary high-resolution polysomnography would help establish that the recurrence signal is not an artefact of the legacy hardware.

The present analyses also do not include a phase-randomised surrogate control: replacing the EEG with surrogate signals that preserve the second-order spectrum but destroy higher-order phase structure would directly quantify the contribution of nonlinear structure beyond the power spectrum to the recurrence metric, and is left for future work.

4.8 Implications for sleep staging and clinical applications

K_0 is a single-channel feature with no fitted constants or learned parameters and no reliance on oscillatory templates or hand-tuned spectral bands — the embedding and preprocessing settings are fixed by convention and shown to be robust across the tested grid (Fig. 1; Supplementary S2, S3) — making it a natural candidate to complement the spectral feature banks used in automated sleep-staging systems. The single-channel feasibility of K_0 has practical relevance beyond methodological tractability: it makes the metric portable to wearable, ambulatory, and bedside monitoring settings — including consumer EEG headbands, home sleep tests, and ICU monitoring of critically ill or minimally responsive patients — where full polysomnography is impractical or contraindicated. The combined feature set reached pooled out-of-subject AUC 0.914 ([0.903, 0.926]) against pooled Wake, decomposing into 0.873 against quiet wake and 0.956 against active-ocular wake, with a small but statistically reliable gain over band power alone ($\Delta\text{AUC} = +0.003$ to $+0.004$, paired-bootstrap $p < 10^{-3}$). The REM-versus-Wake contrast remains the hardest in single-channel pipelines because of the shared low-amplitude, broadband signature; the incremental binomial GLM further suggests K_0 is particularly informative against active-ocular wake ($\Delta\text{AIC} = -744.6$), a common failure mode of automated scoring at sleep onset and at the wake-REM transition. More speculatively, the framework may also be useful for state characterisation in clinical populations whose phenomenologies overlap with REM-sleep behaviour disorder, narcolepsy with cataplexy, certain disorders of consciousness—though the present cross-sectional, healthy-cohort data do not constitute evidence for clinical deployment. The consistent out-of-subject performance and parameter-sparse construction make K_0 a reasonable candidate feature for prospective evaluation in such applied settings.

5 Conclusion

K_0 — a within-subject z-scored summary of one-dimensional persistent homology of delay-embedded single-channel EEG — satisfies the construct-validity criteria of an organisational EEG descriptor of sleep state: it varies systematically across sleep stages, replicates in the independent Sleep-Telemetry sub-cohort, is anatomically gradiated (largest at the frontal midline; preserved in attenuated form on the posterior Pz-Oz derivation), is statistically independent of spectral content (within-subject Pearson r between -0.25 and -0.49 with the four log band powers; opposite REM-Wake direction), is robust across the tested $m \times \tau$ embedding-parameter grid, attenuates with age in a manner consistent with documented age-related changes in REM organisation, and is invariant under a within-subject benzodiazepine challenge. Within-subject normalisation isolated these state-dependent organisational effects from substantial inter-individual variability in raw persistence magnitudes.

On this construct-validity scaffold, the metric also produces a large within-subject REM-

Wake separation (+1.01 SD; Cohen’s $d_z = 1.93$; Holm $p = 2.62 \times 10^{-182}$), with N1 elevated to a level similar to REM and smaller elevations at N2 and N3; the underlying stage means are therefore non-monotonic across the conventional $W \rightarrow N1 \rightarrow N2 \rightarrow N3 \rightarrow \text{REM}$ ordering, with peaks at REM and N1. The N1 finding should be interpreted with caution: N1 is a brief transitional state that contributes relatively few epochs per night for many subjects, and N1 scoring is known to have lower inter-rater agreement than the other stages, so the elevated K_0 in N1 may partly reflect sampling variance and stage-boundary scoring effects rather than a stable internally-sustained-dynamics signature. The REM–Wake separation is preserved across all twelve $m \times \tau$ embedding-parameter combinations tested, demonstrating that the contrast reflects stable geometric properties of the reconstructed state space rather than tuning-specific artefacts.

Conventional EEG spectral, entropy, and complexity measures—including delta–beta band power, permutation entropy, spectral entropy, and Lempel–Ziv complexity—also distinguished REM sleep from wakefulness in this dataset, but in directions that reflect spectral content (e.g. reduced delta and beta power in REM) rather than the recurrence-based elevation captured by K_0 . In leave-one-subject-out classification, band power outperformed K_0 alone in both regimes tested (regime-specific pooled AUCs 0.87 / 0.95 for band power vs 0.70 / 0.84 for K_0 against quiet / active-ocular wake); adding K_0 to band power yielded a small but statistically reliable AUC gain ($\Delta\text{AUC} = +0.004$, paired-bootstrap $p < 10^{-3}$ in each regime). An incremental binomial generalised linear model confirmed that K_0 contributes information beyond band power, with the largest effect for REM versus active-ocular wake ($\Delta\text{AIC} = -744.6$, $p = 2.24 \times 10^{-164}$) and a smaller but still highly significant effect for REM versus quiet wake ($\Delta\text{AIC} = -79.7$, likelihood-ratio $p = 1.62 \times 10^{-19}$). I therefore frame K_0 as a complementary, topologically distinct descriptor of sleep-stage organisation rather than as a superior single-feature classifier.

Collectively, these findings establish topological recurrence (K_0) as a robust, replicable descriptor of sleep-stage organisation that captures geometric structure of the reconstructed neural state space not directly indexed by frequency-based or signal-complexity measures. Independent-cohort replication on the Sleep-EDF Telemetry sub-cohort, posterior-electrode generalisation to Pz–Oz, EOG-corrected and EOG-channel artefact controls, and leave-one-subject-out classification at AUC = 0.95 (combined feature set) jointly support the use of K_0 as a complement to standard spectral analyses for identifying brain states dominated by internally sustained activity. The framework is observational and descriptive: claims are restricted to statistical associations between K_0 and expert-scored sleep stages in a publicly available polysomnography dataset.

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Author contributions

Emmett McMahon: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization.

Declaration of competing interests

The author declares that there are no known financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics Statement

The analyses in this study were performed on the Sleep-EDF Expanded dataset, a publicly available collection of polysomnographic recordings distributed through PhysioNet. All recordings were previously collected with informed consent and ethical approval obtained by the original investigators. The dataset contains only de-identified physiological recordings and associated sleep stage annotations. As this study involved secondary analysis of publicly available anonymised data and did not involve direct contact with human participants, additional institutional ethics approval was not required. All analyses complied with the data usage guidelines provided by PhysioNet.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used a generative AI tool to assist with exploratory discussion and clarification of mathematical concepts and notation, as well as to support language editing and clarity of presentation. The AI tool was not used to generate results, perform data analysis, or derive scientific conclusions. All mathematical formulations, analyses, interpretations, and final written content were developed, verified, and are the sole responsibility of the author.

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Code Availability

The full analysis pipeline used to produce every result reported in this manuscript — including the main TDA computation, all robustness and sensitivity analyses, the demographic stratification, the power analysis, and the supplementary round-3 and round-4 analyses — is publicly available and runs from a single command. The code is hosted at <https://github.com/mcmahonemmet/Topological-Recurrence-in-EEG-Dynamics> and a permanent, version-tagged release has been pre-deposited on Zenodo prior to manuscript submission (DOI: <https://doi.org/10.5281/zenodo.18863540>). Running `pythonrun.py` from a fresh checkout executes the complete analysis end-to-end in four stages: the consolidated `pipeline.py` (main TDA, baselines, robustness grid, and all mixed-effects statistics); `collate.py` (assembling

999 every result into a single workbook); `figures.py` (all manuscript and supplementary fig-
1000 ures); and the five standalone supplementary-analysis scripts (`demographics_breakdown.py`,
1001 `stratified_effects.py`, `sensitivity_power_analysis.py`, `supplementary_round3.py`,
1002 `supplementary_round4.py`). The repository also includes the frozen requirements file. Re-
1003 running `run.py` from a fresh checkout reproduces every output CSV and figure reported in this
1004 manuscript.